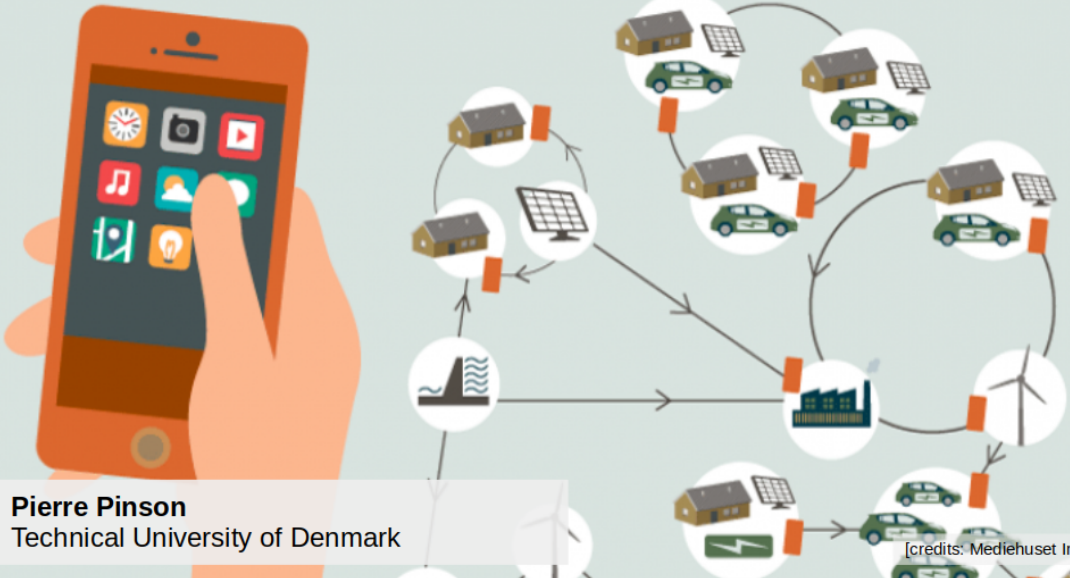
Module 2 – Electricity Spot Markets (e.g. day-ahead)

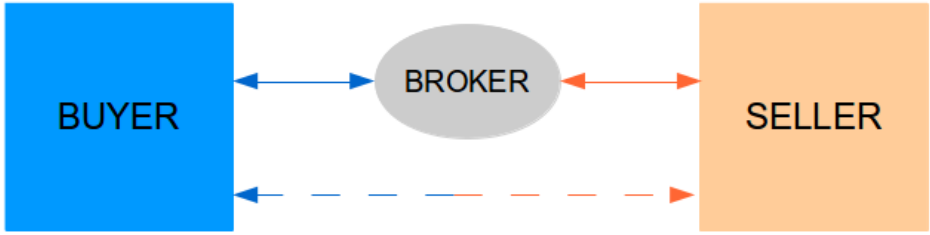
2.1 From bilateral contracts to the electricity pool



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Bilateral contracts



- **Bilateral contracts** are for a *direct exchange* of energy between a buyer and a seller, in a decentralized fashion
- They may both be producers and/or consumers
- Most likely a broker is involved...
- Eventually, the *system operator is informed* about the trades that occurred

Types of bilateral trading

- **Customized long-term contracts**:
  - very flexible contracts (basically, you can try to negotiate whatever you want)
  - private transactions (conditions are fully unknown to others)
  - large transactions costs
  - large amounts of energy, over long periods of times

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- **Over the counter (OTC) trading**:
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  - typically, smaller amount and short lead times
- **Electronic trading**:
  - based on an electronic platform that consistently match supply and offer bids
  - virtually no transactions costs
  - very fast, therefore allowing trading "until the last second"

Placing it into perspective...

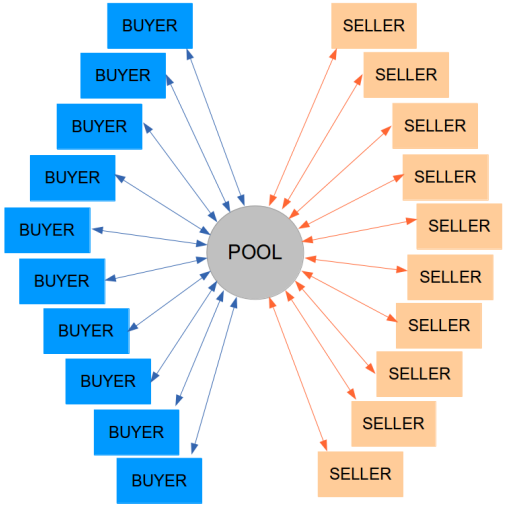
- Bilateral trading may be interesting...
- but the pool provides **a centralized form of system management**, which seems to be increasingly preferred in Europe for day-ahead markets.

Example:

- Nord Pool Spot is the Europe's largest power market
- 505 TWh of energy traded in 2016
- Nordic and Baltic day-ahead auction Elspot represents 391 TWh of energy traded
- Average system price of 26.91€
- In Elspot: 380 buyers/sellers - >2000 orders a day

- Let us focus on pools and auctions for now...

Auctions in an electricity pool



- All generation bids and consumption offers are placed **at the same time**
- **No-one knows about others' bids and offers**
- A centralized *market-clearing algorithm* decides about bids and offers that are retained
- Eventually, the *system operator* is informed about the trades that occurred

An example auction setup

- *Deadline for offers*: 29th of January, 12:00 - *Delivery period*: 30th of January, 11:00-12:00
- Supply and demand offers include:

Demand: (for a total of 1065 MWh)

Company	Supply/Demand	id	Amount (MWh)	Price (€/MWh)
CleanRetail	Demand	D_1	250	200
El4You	Demand	D_2	300	110
EVcharge	Demand	D_3	120	100
QualiWatt	Demand	D_4	80	90
IntelliWatt	Demand	D_5	40	85
El4You	Demand	D_6	70	75
CleanRetail	Demand	D_7	60	65
IntelliWatt	Demand	D_8	45	40
QualiWatt	Demand	D_9	30	38
IntelliWatt	Demand	D_{10}	35	31
CleanRetail	Demand	D_{11}	25	24
El4You	Demand	D_{12}	10	16

An example auction setup

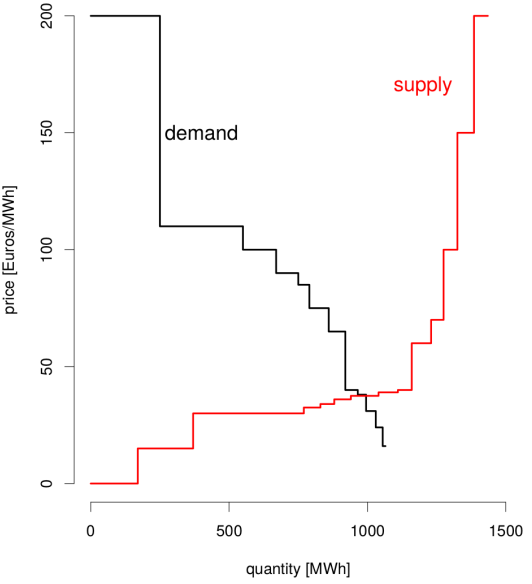
Supply: (for a total of 1435 MWh)

Company	Supply/Demand	id	Amount (MWh)	Price (€/MWh)
RT®	Supply	G_1	120	0
WeTrustInWind	Supply	G_2	50	0
BlueHydro	Supply	G_3	200	15
RT®	Supply	G_4	400	30
KøbenhavnCHP	Supply	G_5	60	32.5
KøbenhavnCHP	Supply	G_6	50	34
KøbenhavnCHP	Supply	G_7	60	36
DirtyPower	Supply	G_8	100	37.5
DirtyPower	Supply	G_9	70	39
DirtyPower	Supply	G_{10}	50	40
RT®	Supply	G_{11}	70	60
RT®	Supply	G_{12}	45	70
SafePeak	Supply	G_{13}	50	100
SafePeak	Supply	G_{14}	60	150
SafePeak	Supply	G_{15}	50	200

That is a lot of offers to match... but how?

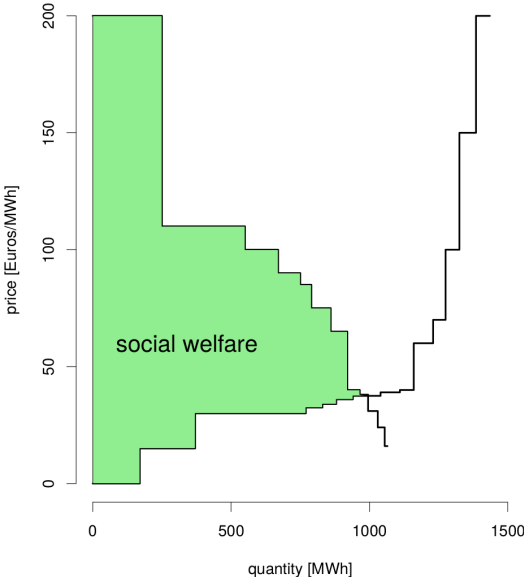
Merit order and equilibrium

- Consumption offers are ranked in *decreasing price order*
- Supply offers are ranked in *increasing price order*
- This defines the **merit order**
- A "magic" point appears: the equilibrium point between supply and demand...



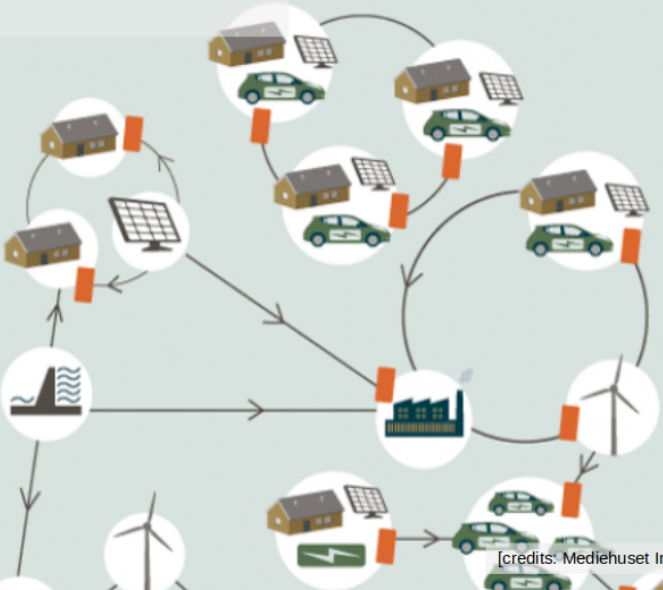
Social welfare and its maximization

- *Social welfare is defined as the area between consumption and generation*
- This equilibrium point is that which allows to maximize social welfare
- *Why?*
  - **Any buyer is to pay at most what he was ready to pay**
  - **Any seller will get at minimum the price he was ready to sell for**



Use the self-assessment quizz to check your understanding!





[credits: Mediehuset Ingeniøren]